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CityFly: Signmark-guided autonomous UAV navigation with vision language model☆

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ABSTRACT

Vision-Language Navigation (VLN) aims to enable agents to integrate visual perception information while receiving human language instructions to complete navigation and task execution, with broad application prospects. In recent years, there have been related explorations in scenarios such as urban air mobility, but most existing research focuses on aerial perspective modeling; in practical low-altitude applications, such as drone last-mile delivery in the context of the low-altitude economy, drones often need to navigate through low-altitude street spaces and accomplish complex tasks. In addition, traditional navigation methods, whether ground-based or aerial, usually rely on a few prominent landmarks for positioning and decision-making. However, in most urban environments, prominent landmarks are scarce, whereas readable signmarks (a structured integration of textual, symbolic, and graphic elements that aim to convey distinctive semantics) are denser and more stable, offering richer and practical navigation cues. Based on the aforementioned issues, this paper proposes a signmark-guided drone vision-language navigation (SG-VLN) task for low-altitude urban street scenarios. It also builds a simulation environment platform that is closer to the real world, compatible with both 3D maps and 3DGS types, while providing a more convenient method for automatic trajectory generation. On this platform, we built a signmark-based VLN dataset, containing 387 short trajectory tasks with an average length of about 130 m, and 1710 long trajectory tasks with an average length of about 342 m. For the proposed signmark-guided VLN tasks, this paper further proposes the signmark-guided agent model. Compared to previous models, we have added a human verbal description language conversion module; introduced a historical trajectory information record mechanism that integrates multiple recent image frames; and designed a fine-grained sub-trajectory modeling strategy, thereby improving navigation performance and robustness in complex low-altitude street environments.

1. Introduction

Vision-Language Navigation (VLN) [1] represents a critical research direction in embodied intelligence, aiming to enable intelligent agents to perform goal-directed navigation decisions based on natural language instructions combined with visual observations. In recent years, as related research has continuously expanded from indoor scenarios [2] to continuous environments [3], real-world street views [4], and more open three-dimensional spaces [5], the application targets of vision-language navigation have gradually extended from traditional ground-based agents to unmanned aerial vehicle platforms. Compared to ground-based scenarios, UAVs possess broader spatial coverage and more flexible maneuverability in urban low-altitude environments, making autonomous navigation research in open and complex scenarios more practically valuable. Meanwhile, with the development of the low-altitude economy and the continuous growth of urban low-altitude application demands, UAVs have been widely deployed

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in tasks such as urban inspection [6], emergency response [7], logistics delivery [8], and infrastructure monitoring [9]. These applications not only require systems to possess large-scale, time-critical autonomous operation capabilities, but also impose higher demands on low-barrier human–machine interaction and highly reliable online navigation. Therefore, UAV vision-language navigation oriented toward low-altitude urban street scenes not only has clear application prospects, but also presents new research challenges for multi-modal perception, scene understanding, and sequential decision-making.

Despite the broad application prospects, current research on UAV VLN still exhibits significant gaps in scene settings and environmental adaptation. On one hand, many existing studies predominantly focus on top-down [10] or bird's-eye view navigation paradigms [11]. However, in real-world low-altitude economic applications, UAVs frequently need to shuttle through complex urban streets to execute precise point-to-point tasks (such as accurate cargo delivery). This specific "low-altitude shuttling" requirement has yet to be systematically characterized and thoroughly explored. Furthermore, while maps [12] and GPS guidance might seem reliable for aerial tasks, they frequently encounter issues in complex urban canyons, such as untimely map acquisition and updates, positioning errors, and signal occlusion. Consequently, developing autonomous visual navigation capabilities for UAVs operating under map-less or weak-positioning conditions is imperative.

Operating in these complex street environments requires reliable visual anchors for navigation, yet current methods face limitations in information utilization. Many approaches rely on prominent landmarks [13] to specify tasks and guide paths (e.g., aligning instructions via descriptions of the landmarks' color, shape, and material). However, in actual urban environments, such prominent landmarks are relatively scarce, and many areas consist of highly repetitive, indistinct building clusters [14], making this approach difficult to generalize. Another class of methods, represented by NavAgent [15], utilizes vision-language alignment models like CLIP to retrieve common street objects (e.g., traffic lights, trash bins) as navigation anchors. While this reduces reliance on prominent landmarks, it heavily focuses on visual appearance matching. This deviates from natural human verbal instructions, which typically emphasize destinations [16] and action intentions rather than detailing the physical appearance attributes of common objects.

Crucially, existing research often overlooks the most ubiquitous and direct source of guidance information [17] in urban environments: signmarks (a structured integration of textual, symbolic, and graphic elements that aim to convey distinctive semantics, such as shop, traffic, and public guidance signs). Although some studies have implicitly involved similar cues, they generally remain at the superficial level of having similar prompts, failing to utilize the explicit content of the signmark as the core semantic basis for driving decision-making. In human cognition, places are much easier to remember and identify by their signmarks, and human navigators naturally prefer signmark-based expressions [18] when giving instructions. To bridge this gap, we propose the signmark-guided UAV vision-language navigation (SG-VLN) task for low-altitude urban street scenarios. This task aims to better align with real-world application needs and systematically explore cross-modal understanding and autonomous navigation capabilities driven explicitly by signmark content.

The SG-VLN task focuses on completing navigation by relying on the vast amount of signmark information found throughout urban streets as guidance. Achieving this goal involves three key components: First, building a low-altitude urban street simulation data generation platform that closely resembles the real world, so that the distribution, visibility changes of signmark, and block structures better meet practical application needs; Second, establishing a supporting task and data system around signmark guidance, providing datasets that cover trajectories of different route scales and difficulty levels, ranging from short to long distances; Third, designing a navigation model for this task that incorporates a historical trajectory information recording mechanism, integrates image frames from multiple recent moments, and develops fine-grained sub-trajectory modeling strategies, thereby enhancing navigation performance and robustness in complex low-altitude street environments.

Compared to previous data generation platforms that mainly focus on indoor environments, existing data generation platforms for urban aerial scenarios [19] still exhibit a mismatch with real-world application needs: on the one hand, they often rely on conceptual virtual cities like BigCity [20], with relatively simplified scene structures, insufficient restoration of real urban details, and a lack of a key element—the abundant signmark information present throughout streets and alleys; on the other hand, the point cloud-based map representation provided by CityNav makes it difficult to present clear and readable signmark content, further limiting the usability of signmark cues in navigation. To address these shortcomings, this paper constructs a unified urban low-altitude street simulation data generation platform: part of it is based on the UE5 engine combined with the Cesium plugin, using publicly available city maps of the Hong Kong area for data generation; another part introduces 3DGS technology to utilize self-collected urban street-view data for testing and supplementary validation, thereby achieving more realistic urban scene appearances and clearer signmark presentation, while also supporting automated trajectory generation and annotation, lowering the barrier for researchers to expand and reproduce data.

Based on this platform, this paper further constructs a SG-VLN dataset, making the task more aligned with real urban navigation needs: it provides more detailed definitions for possible street scenarios, requires the drone agent to execute higher-level action instructions (such as "turn right at the next intersection"), and recognize more complex and variable signmark information in street scenes; at the same time, it offers diversified settings for short and long trajectories and provides higher-resolution images to improve signmark readability, facilitating the model to fully utilize signmark cues for decision-making.

At the model level, this paper proposes the signmark-guided Agent: based on the Qwen vision-language large model, it further introduces three key enhancements compared to previous methods—(i) a spoken language conversion port, designed to transform human oral descriptive language into standard navigation language; (ii) the fusion of multiple recent image frames, to enhance temporal perception and improve stability in situations with occlusion or changes in viewpoint; (iii) a fine-grained sub-trajectory modeling strategy oriented toward signmark cues, which requires the model to maintain a sub-trajectory linked list and explicitly determine whether a certain key signmark has been seen, and whether a corresponding stage has been entered, thereby improving navigation performance and robustness in low-altitude complex street environments.

The contributions of this work are four-fold:

1. Task: We formulate a signmark-guided low-altitude street-level drone VLN problem leveraging dense urban signmarks.
2. Platform: We build a high-fidelity, reproducible simulation and data-generation platform for low-altitude urban streets.
3. Dataset: We construct a signmark-based VLN dataset aligning instructions with in-scene signmark cues.
4. Model: We propose the signmark-guided agent with colloquial rewrite, multi-frames fusion, and sub-trajectory modeling for robust navigation.

2. Related work

2.1. Visual language navigation

VLN research can generally be classified into three main application scenarios: indoor ground, outdoor ground, and aerial. Overall, the evolution of methods is shifting from early task-specific models to a unified

Fig. 1. Comparison between landmarks and signmarks. Signmarks provide denser and more distinctive guidance than landmarks. We also counted the average number of references in the scenes, and there are more signmarks.

representation and reasoning paradigm centered on vision-language foundation models (VLMs), progressively expanding toward navigation requirements involving longer timeframes and complex trajectories.

In the realm of indoor ground robot goal-oriented navigation, early benchmarks such as R2R [1] and RxR [2] laid the foundation for mapping natural language to discrete viewpoints, while VLN-CE [3] pushed the boundary toward continuous 3D environments. To enhance autonomous decision-making in these spaces, recent methods focus on advanced spatial reasoning. For instance, VL-Nav [21] emphasizes real-time spatial reasoning to handle dynamic obstacles. More comprehensively, WMNav [22] integrates VLMs into a world model framework, providing feedback for policies by predicting action outcomes and building world model memory. It maintains an online Curiosity Value Map to dynamically quantify the likelihood of a target's presence, adopting a two-stage action proposal (broad exploration first, precise localization later) to mitigate decision biases caused by model hallucinations. However, indoor VLN methods are inherently constrained by limited spatial scales, structured room layouts, and predefined navigation graphs, lacking the environmental complexity and scale required for outdoor open-world applications.

Stepping outdoors, urban street-view VLN introduces complex visual elements and longer trajectories. Foundational works like Touchdown [4] established paradigms for natural language navigation and spatial reasoning in real street environments. To tackle the increasing complexity of urban scenes, NavAgent [15] attempts to fuse multi-scale urban street views to enhance embodied navigation. Pushing the capability of foundation models, FLAME [23] proposes a Flamingo-based multimodal LLM navigation agent specifically for street-view tasks. It emphasizes the efficient integration of multi-time observations and enhances modeling capability for complex street-level instructions through a three-stage fine-tuning paradigm combined with automatically synthesized data. Furthermore, researchers are proactively constructing platforms and benchmarks specifically targeted at long-horizon vision-language navigation [24] to address the severe memory degradation over extended paths. Despite these advancements in understanding complex street instructions, ground-based outdoor navigation methods are fundamentally restricted by 2D road networks and traffic topologies, lacking the three-dimensional mobility and altitude flexibility required for emerging aerial applications.

In aerial top-down drone VLN, FlightGPT [25] addresses issues of insufficient multimodal fusion, weak generalization, and poor interpretability by constructing a two-stage VLM-based training process (high-quality demonstration SFT warm-up + RL/GRPO optimization based on composite rewards) and introduces a Chain of Thought reasoning mechanism to improve policy transparency and cross-scenario robustness. Additionally, to improve human–machine collaboration in the air, recent studies have begun exploring fine-grained alignment learning for aerial vision-dialog navigation [26].

2.2. Landmark-based UAV vision-language navigation

Early studies published explored stochastic reasoning for structural pattern recognition in image-based UAV navigation [27], laying the foundational theories for matching geometric targets. In existing research on UAV visual-language navigation, landmarks are often considered the primary environmental cues for supporting target localization and path planning. For example, OpenFly [28] focuses on building a landmark-centered data generation and task platform, providing a comprehensive platform and large-scale benchmarks for aerial VLN, with an automated process that identifies and selects landmarks as navigation markers. From a top-down/macroscopic perspective, UAV-VLPA* [5] generates task-level flight paths using satellite images and language instructions, combining TSP (global waypoint sequencing) and A* (local obstacle avoidance) for planning; CityNav extracts target/landmark/environment names from language and grounds them in observations using detection and segmentation, dynamically building maps for decision-making; FlightGPT [25] is evaluated on CityNav and focuses on city-scale goal understanding and localization to enhance generalization and interpretability — these setups overall rely more on a small number of stable and recognizable landmarks to support position judgment and navigation, exhibiting task characteristics similar to visual localization. In near-ground low-altitude urban scenarios, the instructions and trajectories of AerialVLN [13] are organized around predefined landmarks, prompting the model to learn the alignment between language references, visual correspondences, and action decisions. NavAgent [15] goes a step further by treating the difficulty of matching fine-grained landmarks as a core challenge and proposes a multi-scale fusion framework: landmark extraction, GLIP landmark recognition, topological graph encoding, and LLM decision-making, while also developing NavAgent-Landmark2K to enhance landmark recognition capabilities. Meanwhile, other studies further explicitly incorporate landmarks into method design, strengthening cross-modal alignment and decision-making reliability through landmark matching, landmark graph representation, or entity-landmark alignment mechanisms, thereby improving navigability and robustness in complex urban environments.

However, the above-mentioned "landmark-based" routes share a common shortcoming: they often underplay the abundant signmarks [29] present in low-altitude urban environments, which is crucial for human navigation (such as store signs, street signs, directional signs, numbers, and warning texts [30]; see Fig. 1, where landmarks are sparser and more common, while signmarks are denser and more distinctive). As shown in the figure, the average quantity of signmarks significantly exceeds that of landmarks, highlighting the importance of signmarks in low-altitude urban environments. From the perspective of human cognition and practical usability, signmark-based navigational cues are generally easier to recognize and verify than landmarks that

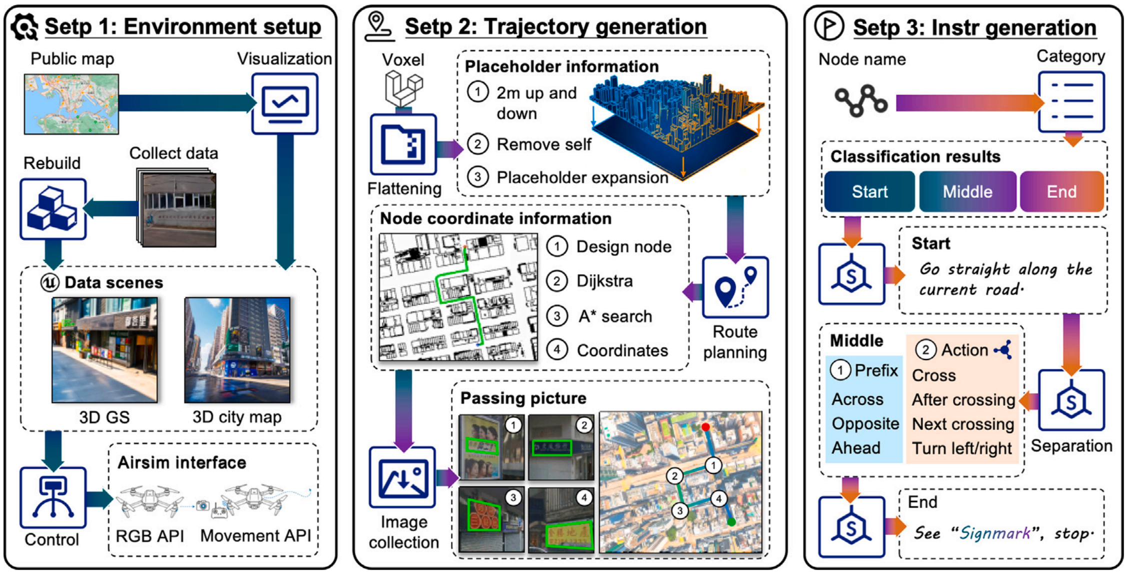

Fig. 2. Data generation platform overview: (i) **Environment:** UE5 + Cesium Hong Kong 3D city map, plus 3DGS street-view test scenes for data collection and AirSim control. (ii) **Trajectory:** voxel-to-2D grid, Dijkstra for global planning, A* for local paths, image collection. (iii) **Instruction:** extract key events and generate colloquial "start–middle–end" instructions to form SG-VLN instruction–trajectory pairs.

are visible from a distance but vary in appearance, and they provide more direct decision-making signals in local tasks (such as turning, arriving, or confirming a position). Therefore, the following sections will further compare navigation settings based on "landmarks only", "signmarks only", and "a combination of both", and experiments will be conducted to verify that incorporating signmark in low-altitude urban scenarios can lead to more reasonable and stable navigation outcomes.

3. Signmark-guided visual language navigation task

We propose a signmark-guided visual-language navigation task, SG-VLN, aimed at low-altitude street scenes, designed to guide drones to perform goal-oriented navigation between continuous street views and shop facades. This task is carried out without maps or GPS/localization priors: during flight, the drone only receives pre-given spoken navigation instructions from humans and, based on the history of collected images and the current image, predicts the next discrete action online. This allows the drone to follow each stage of the signmark guidance without collisions, reaching the vicinity in front of the target shop and stopping.

Formally, given a natural-language spoken instruction $\mathcal{I} = (i_1, \dots, i_N)$, where i_k denotes a stage of SG-VLN, the UAV receives at time t a visual observation as an RGB image $O_t \in \mathbb{R}^{H\times W\times 3}$, and its observation history is $O_{1:t} = (O_1, \dots, O_{t-1})$. At each time step, the agent needs to understand its current location and whether it observes the guidance signmarks, and predicts the next action a_t based on the input:

$$a_t = f(\mathcal{I}, O_{1:t-1}, O_t), \qquad a_t \in \mathcal{A}. \tag{1}$$

The action space is a fixed discrete set:

$$\mathcal{A} = \{stop, forward3\ m, turnleft15^\circ, turnright15^\circ\}. \tag{2}$$

At each step, the agent selects exactly one action from the above set and executes it once; compared with OpenFly, we use finer-grained rotation angles.

The success condition is defined as follows: the UAV chooses *stop* at some terminal time T, and the Euclidean distance between its position p_T and the target shop entrance location g satisfies $\|p_T - g\|_2 \le 6$ m. This 6 m success radius is stricter than the 20 m threshold used in OpenFly, thus imposing higher requirements on fine-grained decision making and vision–language alignment in low-altitude street scenes.

4. Data generation platform

In order to accomplish the SG-VLN task, we need more realistic virtual environments to generate training data and test environments to support its data requirements. As shown in Fig. 2, this is the overall process.

4.1. Environment setup

First, we selected a publicly available 3D map of Hong Kong as the main environment. This map contains a large number of readable signmark elements at the street level and maintains a high level of clarity even from low-altitude street-level views. In contrast, some 3D city data may look acceptable from a high-angle aerial view but often suffer from insufficient texture resolution and unreadable signmark at close-range or street-level perspectives, making them inadequate for SG-VLN's requirement for "readable signmarks". From an engineering perspective, directly importing the entire city model as a static mesh into the engine can cause two main problems: first, a sharp increase in GPU memory usage, resulting in high operational and editing costs; second, potential loss of detail during processes such as conversion, batching, and texture handling, further reducing the texture clarity at street-level perspectives. To address this, we adopted Cesium's streaming loading solution: instead of baking the entire city scene as a local static mesh at once, we use a plugin to load and render the model in chunks from disk (or other data sources) on demand during runtime/editing. This approach significantly reduces one-time resource usage while maintaining interactivity and better preserves close-range texture details and signmark readability. After completing the streaming load and coordinate setup, we deployed the scene into a UE5 level and further configured modules such as drone sensors, colliders, lighting, and post-processing within the level to achieve a high-quality simulation environment suitable for trajectory generation and data collection. This environment construction echoes prior remote-sensing studies on urban road understanding, which emphasize preserving road structure and scene organization from visual data [31], and on polygonal building contour extraction [32], which highlights retaining fine-grained building facades—exactly where street-level signmarks reside.

Although city-level maps provide a macroscopic environment close to real urban structures, gaps remain in material realism, fine-grained geometry, local occlusions, and lighting. To test robustness under more realistic imaging, we additionally built several 3D GS test environments: we captured the target area with 360° panoramic cameras for wider coverage, then applied standardized preprocessing before 3D GS reconstruction and optimization. This yields renderable scenes with richer geometry, finer appearance, and stronger local realism, supporting both qualitative and quantitative evaluation under high-precision visual conditions.

To achieve cross-scenario, reusable drone control and data collection, we use AirSim [33] as a unified simulation and control interface, abstracting the drones in UE5 as programmable agents. AirSim provides standardized sensor access and motion control capabilities, allowing the same control logic to run seamlessly across different levels/environments (such as the Hong Kong urban scene and the 3DGS test scene), thereby ensuring consistency and scalability in the data generation process. Similar to how recent works like Autobio [34] established high-fidelity simulation benchmarks for robotic automation in digital laboratories, our platform provides a standardized, reproducible digital twin dedicated to low-altitude UAVs.

4.2. Trajectory generation

To automatically generate executable navigation trajectories, we first query the scene's 3D occupancy/voxel map via the AirSim interface and slice it at the UAV flight altitude h to obtain a 2D traversable grid M_h (free/occupied). This abstraction of traversable road structures is inspired by prior remote-sensing studies on road understanding, which highlight the importance of continuity and connectivity in representing road layouts from visual imagery [35]. We then define two semantic node sets—intersections $\mathcal{V}_{int}$ and shops $\mathcal{V}_{shop}$—and build an intersection graph $G = (\mathcal{V}_{int}, \mathcal{E})$, where each edge is weighted by planar distance or traversability cost between intersections. This graph construction is also inspired by remote-sensing studies on road understanding, which suggest that representing road scenes with structured semantic elements can provide a useful perspective for organizing complex road layouts [36].

Given a start shop $s \in \mathcal{V}_{shop}$ and a target shop $t \in \mathcal{V}_{shop}$, we select the closest intersection nodes $v_s, v_t \in \mathcal{V}_{int}$ (in Euclidean distance) as the entry and exit for global planning. We then compute the minimum-cost intersection path on G using Dijkstra's algorithm:

$$\pi^* = \arg \min_{\pi \in \Pi(v_s, v_t)} \sum_{e \in \pi} w(e), \tag{3}$$

where $\Pi(v_s, v_t)$ denotes the set of feasible paths from v_s to v_t on G, and $w(e)$ is the edge weight.

For each adjacent intersection pair (v_i, v_{i+1}) along π^*, we perform local planning by running A* search on the 2D grid M_h to obtain a continuous pixel-level (or meter-level) trajectory segment. Concatenating all local segments yields the complete trajectory τ. The platform also supports manually adding, removing, or editing intersection/shop anchors to generate routes with specific topology or difficulty, enabling diverse data sampling.

4.3. Instruction generation

After obtaining a trajectory τ, we further convert it into a sequence of colloquial navigation instructions for SG-VLN. Different from prior work that treats the existence of signmark as a weak cue, we explicitly use readable street signmark as a semantic anchor and organize both decision points and instruction templates around it.

Specifically, we first perform key-event mining along the trajectory τ and identify a set of decision points $\mathcal{D}$ associated with semantic nodes, including intersection turning points and action-trigger points for crossing roads/crosswalks; for each trigger point, we select a corresponding signmark (denoted as m_k). We then organize the navigation into a structured instruction sequence $\mathcal{I} = \{i_1, \dots, i_N\}$, where each instruction is written using a three-part "start–middle–end" template to ensure coherence and readability.

The start segment describes initial behaviors such as going straight or following the road. The middle segment is triggered when a signmark m_k is detected, and each instruction is instantiated as:

$$i_k : \text{ When you see } m_k, \; p + a, \tag{4}$$

where the optional prefix p is selected from $\mathcal{P} = \{across, prefix, ahead\}$ to indicate the target's relative position with respect to the landmark, and the action a is selected from an action-template set and bound to the trajectory via a parameterized slot (*left/right*).

The end segment describes the stopping condition near the destination. In this way, geometric constraints (turning direction, intersection order, and whether to cross the street) and perceptible semantics (signmarks) are jointly mapped to natural-language instructions, forming paired instruction–trajectory data for the navigation task.

5. Dataset

To systematically evaluate SG-VLN capabilities, we constructed the SG-VLN dataset on the aforementioned platform. Unlike general VLN benchmarks that primarily focus on large-scale open spaces and sparse salient landmarks, this dataset concentrates on low-altitude urban streets with dense signmarks, emphasizing the information density of single trajectories, the difficulty of language–geometry alignment, and task verifiability.

5.1. Divide

The dataset contains approximately 2.1k trajectory-instruction pairs, including:

Short trajectories: 387 trajectories, with an average path length of 130 m, mainly covering basic tasks such as a single intersection and a single signmark.

Long trajectories: 1710 trajectories, with an average path length of 342 m, usually spanning up to 5–6 intersections, covering roughly four blocks and multiple levels of signmarks.

To fairly evaluate the model's generalization ability, we construct a training set and two types of test sets on 3D Map (a 3D city map of Hong Kong): Seen (regions observed during training) and Unseen (previously unobserved regions). In addition, to examine robustness under imaging conditions closer to real-world scenarios, we further build the Test-Reality-Unseen (3DGS near NWPU) test set, ensuring that its scene is completely excluded during training, thereby enabling a strict cross-domain evaluation. Fig. 3 and Table 1 reports the statistics of each split, including the number of trajectories (Traj), average path length (Avg Len), average number of action steps (Avg Actions), average number of junctions (Junc), and the length range (Len Distrib).

5.2. Instruction and language analysis

The instructions for SG-VLN are designed for navigation scenarios in low-altitude urban streets with dense signmarks, emphasizing the close alignment of observable and verifiable signmarks with the agent's executable actions (such as moving forward, turning, crossing streets/crosswalks, and following roads). This makes the evaluation more focused on the complete chain of reading signmarks—locating signmarks—triggering decisions. Compared to general VLN settings that mainly rely on macro geometric landmarks (such as large building outlines or terrain boundaries), SG-VLN's language more frequently references street-level shop names, sign keywords, and directional/relative position phrases, making each instruction denser in information and imposing stronger requirements for fine-grained alignment between language, vision, and geometry.

As illustrated in Fig. 4(a), Hong Kong urban scenes exhibit a dense and highly cluttered signmark environment at the street level: along a

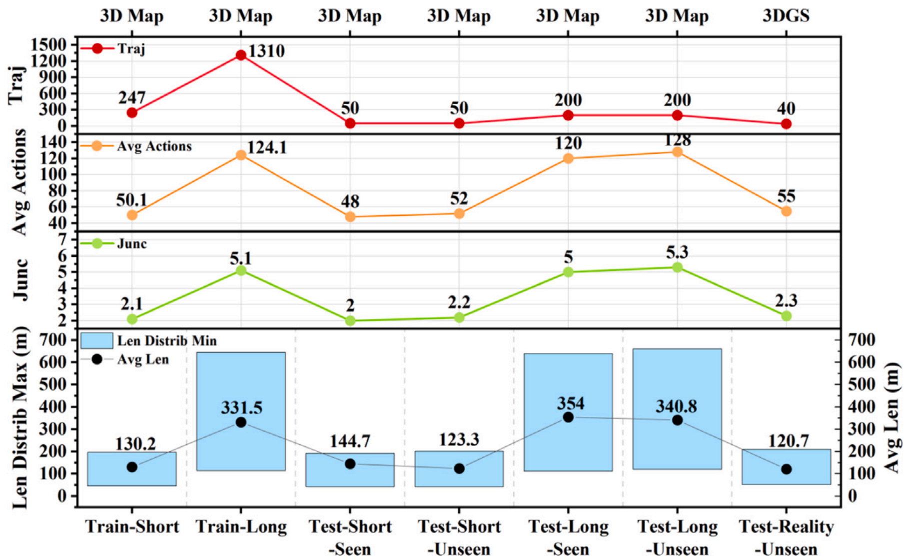

Fig. 3. Dataset splits and statistics. To evaluate generalization, we build training and two test splits on 3D Map: Seen and Unseen, and additionally include Test-Reality-Unseen (3DGS) for strict cross-domain evaluation under more realistic imaging conditions.

Table 1
Dataset splits and statistics.

Split	Scenes	Traj	Avg Len (m)	Avg Actions	Junc	Len Distrib (m)
Train-Short		247	130.2	50.1	2.1	45–197
Train-Long		1310	331.5	124.1	5.1	114–645
Test-Short-Seen		50	144.7	48.0	2.0	42–192
Test-Short-Unseen		50	123.3	52.0	2.2	42–201
Test-Long-Seen		200	354.0	120.0	5.0	111–639
Test-Long-Unseen		200	340.8	128.0	5.3	120–660
Test-Reality-Unseen	3D GS	40	120.7	55.0	2.3	51–210

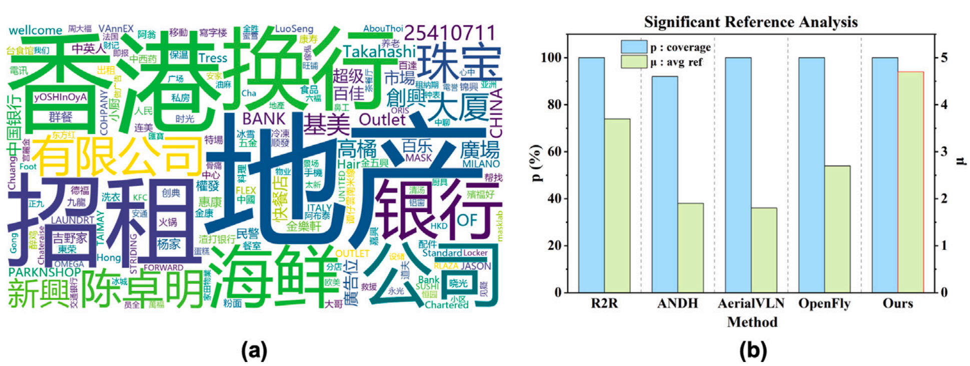

Fig. 4. Instruction and language analysis. (a) Dense street-level text in Hong Kong scenes; (b) Reference statistics across datasets/methods (coverage $p\%$ vs. avg ref μ).

single trajectory segment, numerous Chinese signmark words as well as alphanumeric tokens may appear simultaneously, and such signmarks often co-occur with decision-relevant regions such as intersections, storefront facades, and road boundaries. Therefore, introducing signmark anchor points in the instruction as *verifiable trigger conditions* can improve the interpretability and determinability of navigation, but it also imposes higher requirements on the agent: it must not only perceive signmark robustly, but also disambiguate and correctly align the target signmark among multiple visually similar candidates.

Furthermore, Fig. 4(b) presents comparative results of different datasets [1,13,28,37] regarding the reference phenomenon. We observe that, while maintaining relatively high overall coverage $p(\%)$, different datasets show clear differences on the phenomenon-intensity metric μ, suggesting inconsistency in the explicit usage frequency and density

Table 2
Statistics comparison with prior VLN datasets.

Dataset	Signmark	Ntraj	Path len (m)	Intr len	Action space	Environment
R2R [1]	✗	7.2K	10	29	Graph-based	Matterport3D
RxR [2]	✗	14K	14.9	129	Graph-based	Matterport3D
REVERIE [38]	✗	7K	10	18	Graph-based	Matterport3D
CVDN [39]	✗	7.4K	25	34	Graph-based	Matterport3D
TouchDown [4]	✗	9.3K	313.9	90	Graph-based	Google Street View
VLN-CE [3]	✗	4.4K	11.1	19	2 DoF	Matterport3D
LANI [40]	✗	6K	17.3	57	2 DoF	CHALET
ANDH [37]	✗	6.2K	144.7	89	3 DoF	xView
AerialVLN [13]	✗	8.4K	661.8	83	4 DoF	AirSim + UE
CityNav [19]	✗	32K	545	26	4 DoF	SensatUrban
OpenUAV [41]	✗	12K	255	104	6 DoF	AirSim + UE
OpenFly [28]	✗	100K	99.1	59	4 DoF	AirSim + UE, GTA V, Google Earth Studio, 3DGS + SIBR viewers
SG-VLN (Ours)	✓	2.1K	297.3	94.1	3 DoF	AirSim + UE (3D Map, 3D GS)

of reference cues. In contrast, SG-VLN introduces denser signmark references that more closely resemble real streets, enabling instructions to provide finer-grained and verifiable location cues. Although the dataset previously proposed for AerialVLN had a relatively high average number of references, our analysis shows that these references mainly consist of large and simple landmarks such as bridges and highways, which may lack uniqueness and discriminative power. In complex urban environments, this can easily lead to ambiguity and mismatches, weakening the constraint of instructions on decision-making and affecting the model's learning performance. Therefore, we conducted a new analysis of unique references.

5.3. Compare

Table 2 compares SG-VLN (Ours) with existing datasets in terms of trajectory scale, average path length, average instruction length, action space, and environment configuration. Although Ours contains fewer trajectories, dataset value should not be judged by scale alone: on the one hand, we provide an end-to-end automated data generation platform, so dataset expansion is mainly constrained by sampling time and computational resources, enabling sustained and scalable growth; on the other hand, the statistics in Table 2 show that Ours achieves a medium-to-high level in both average path length and average instruction length, indicating that each sample contains a longer spatiotemporal decision chain and denser linguistic constraints, which is beneficial for systematically evaluating mid-to-long-horizon navigation and language alignment.

In terms of usability, unlike solutions that require deploying and configuring multiple environment sources and runtime interfaces, Ours adopts a unified AirSim + UE5 interface that integrates both city-scale high-fidelity maps and 3DGS scenes under the same control and data-collection pipeline, reducing environment-switching overhead and improving reproducibility. Meanwhile, Ours still has certain limitations in its action-space design, which is also reflected in Table 2; however, the primary goal of this work is to validate the effectiveness and evaluation necessity of *signmark grounding* in low-altitude street-level navigation, and thus we prioritize fine-grained alignment between readable signmarks and decision points, while extending to higher-degree-of-freedom action spaces will be an important direction for future improvement.

6. Signmark-guided agent model

Signmark-guided agent is a VLM-based navigation agent designed to make robust, fine-grained decisions by explicitly grounding its actions in street-level signmarks cues rather than relying solely on distant, appearance-varying visual landmarks, as illustrated in Fig. 5.

Colloquial rewrite. The colloquial instruction rewriting module transforms natural language instructions into structured signmark-action steps. The process consists of:

Extracting action steps: Key actions like “go straight” and “turn right” are identified and structured into actionable steps with trigger conditions (e.g., “turn right at the corner”).

Applying constraints: Environmental constraints, such as visibility of a target or specific landmarks, are integrated to ensure the navigation actions are executable (e.g., “stop when you see the shop”).

Rendering instructions: The processed instructions are rendered into a format suitable for the planner's execution.

Multi-frame visual fusion. Street navigation is strongly sequential: viewpoint changes, occlusions, and dynamic objects can make single-frame signmark recognition unstable. We therefore maintain a short-term history window of the most recent L RGB frames and fuse them into an observation O_t for more stable signmark perception. By fusing multi-frame visual information, the model achieves highly robust feature extraction for signmark content. Compared to specialized scene detectors [42] or instance segmentation algorithms [43], our VLM-based agent exhibits superior generalization capabilities, ensuring the accurate capture of semantic anchors even in cluttered and highly variable urban environments.

Fine-grained sub-trajectory modeling. We decompose the full instruction into a sequence of K sub-goals $\mathcal{G} = \{g_1, \dots, g_K\}$. At time t, we maintain a discrete sub-trajectory state (stage index) $S_t \in \{1, 2, \dots, K\}$, where $S_t = k$ indicates that the agent is currently executing the kth sub-goal.

To align with multimodal perception, let O_t denote the fused observation from the latest L frames. For the setting “each stage corresponds to one target signmark”, we assign a unique target signmark m_k to each stage k, denoting the full set as $\mathcal{M} = \{m_1, \dots, m_K\}$. An example is $m_1 = \texttt{DongFangHong}$, $m_2 = \texttt{KFC}$. In practice, the initial target may be empty, i.e., $m_0 = \varnothing$, in which case the agent follows the road straight ahead by default. Each sub-goal is represented as:

$$g_k = (i_k, m_k, \pi_k), \tag{5}$$

where i_k is the normalized language instruction for stage k, and π_k is the within-stage policy (e.g., “keep going straight”, “turn right then go straight”).

In a binary visibility setting, let $V(m_k; O_t) \in \{0, 1\}$, where $V(m_k; O_t) = 1$ indicates that the target signmark m_k is detected in the observation at time t and 0 otherwise; the trigger predicate is simply $\mathrm{Trig}_k(t) = V(m_k; O_t)$.

The sub-trajectory state update is:

$$S_{t+1} = \begin{cases} \min(S_t + 1,\ K), & V(m_{S_t}; O_t) = 1, \\ S_t, & V(m_{S_t}; O_t) = 0. \end{cases} \tag{6}$$

That is, if the target signmark of the current stage becomes visible, we move to the next stage; otherwise we keep the current stage and

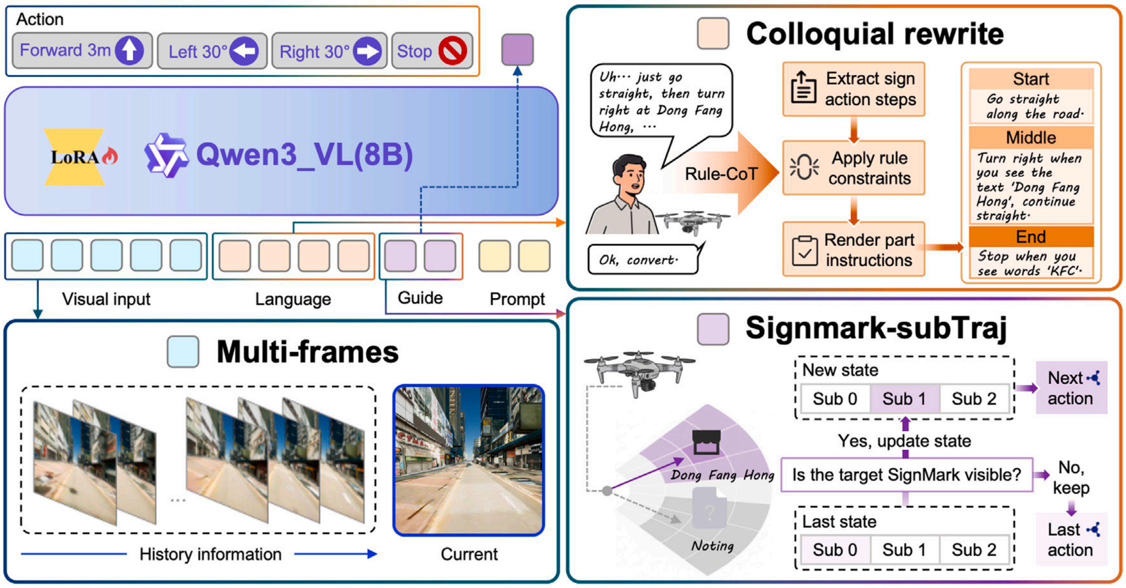

Fig. 5. Signmark-guided agent model. The framework consists of three modules: (i) Colloquial instruction rewriting, which disambiguates user instructions into structured signmark–action steps with explicit trigger/termination conditions; (ii) Multi-frames visual fusion, which fuses the most recent L RGB frames into O_t for stable scene-signmark perception; and (iii) Fine-grained sub-trajectory modeling, which tracks the stage index S_t over K sub-goals and advances when the target signmark m_{S_t} becomes visible.

continue executing its policy. Finally, we write the action decision as a conditional policy:

$$a_t \sim \pi\left(\cdot \mid O_t, i_{S_t} \right), \qquad a_t \in \mathcal{A}, \tag{7}$$

where the action at time t is jointly determined by the fused observation O_t and the current-stage instruction i_{S_t}, forming a closed loop of "stage signmark → stage action → signmark-triggered transition".

Decision and planning. The decision/planning module integrates the fused visual perception, the rewritten structured instruction, and the current stage state to output executable navigation actions while maintaining stage consistency and interpretability.

7. Experiments

In this section, we first introduce the evaluation metrics and experimental setup, and then present the comparison results, the effectiveness study of landmarks and signmarks, and the ablation study.

7.1. Metrics and experimental setup

We use four standard VLN metrics to evaluate different methods, including navigation error (NE), success rate (SR), oracle success rate (OSR), and success weighted by path length (SPL). NE measures the average distance between the UAV's final stopping point and the ground-truth destination. SR is the proportion of successful episodes; an episode is considered successful if the UAV stops within 6 m of the goal. For OSR, an episode is considered successful if any point along the trajectory is within 6 m of the goal. SPL measures success while accounting for efficiency, weighted by the ratio between the ground-truth path length and the actually executed path length.

During fine-tuning, we train on a single multi-GPU server equipped with three NVIDIA GeForce RTX 4090 GPUs (48 GB memory), and use one GPU for inference at test time. Online interactive navigation and simulation rendering are based on Unreal Engine 5, and the simulator runs on a machine with an Intel i5 CPU and an NVIDIA GeForce RTX 5060 GPU (8 GB memory) to ensure real-time rendering, viewpoint updates, and agent interaction.

7.2. Model comparison results

To comprehensively evaluate the proposed method under different navigation distances and cross-scene generalization conditions, we report results on three evaluation splits: *test-short*, *test-long*, and *test-reality-unseen*. For each split, we further divide scenes into *seen* and *unseen* depending on whether the scenes appear during training, in order to systematically examine both in-distribution performance and cross-scene generalization. To ensure fair comparison and reproducibility, we follow OpenFly's experimental setting and include representative baselines such as Random, Seq2Seq, CMA, AerialVLN, Navid, NaVila, and OpenFly, evaluated under the same metrics (NE, SR, OSR, SPL).

Results on *test-short* are reported in Table 3. Results on *test-long* are reported in Table 4. Results on *test-reality-unseen* are reported in Table 5.

Across all three splits (Tables 3, 4, 5), our method achieves the best overall performance, with lower NE and higher SR, OSR, and SPL. The gain mainly comes from explicitly leveraging signmark cues: using target signmarks as stage-transition triggers makes decisions more verifiable and self-correcting, alleviating error accumulation over long horizons. The Qwen-series backbone, trained with signmark detection/recognition capabilities, further stabilizes cue extraction in signmark-rich scenes. In contrast, baselines that do not treat signmark as a key factor—e.g., OpenFly, whose input resolution limits fine-grained signmark representation—suffer in decision-making and localization.

On *test-short* (Table 3), since trajectories are shorter and involve fewer stages, the overall task difficulty is relatively lower, and many methods can obtain reasonable SR/OSR. However, our method still shows higher efficiency and stability on comprehensive metrics such as NE and SPL, indicating that it not only reaches the goal but also tends to complete the task with shorter and more reasonable paths. In contrast, on *test-long* (Table 4), the navigation horizon is longer with more decisions, and errors are more likely to accumulate over time, leading to degraded performance for baseline methods; our method remains leading on this split, suggesting that the signmark-triggered staged execution mechanism is more robust for long-horizon tasks.

Comparing seen and unseen scenes (Tables 3 and 4), unseen scenes usually differ in appearance, layout, or distribution, which causes performance drops relative to seen scenes. Our method follows the same

Table 3
Results on *test-short* (seen/unseen).

Method	test-seen				test-unseen			
	NE↓	SR↑	OSR↑	SPL↑	NE↓	SR↑	OSR↑	SPL↑
Random	122.9 m	0.0%	0.0%	0.0%	110.1 m	0.0%	0.0%	0.0%
Seq2Seq	92.7 m	1.4%	2.8%	0.8%	98.9 m	0.8%	1.9%	0.4%
CMA	78.6 m	3.8%	7.9%	2.0%	88.7 m	2.1%	4.9%	1.0%
AerialVLN [13]	70.9 m	5.9%	11.8%	3.2%	81.6 m	3.2%	7.4%	1.6%
Navid [44]	76.8 m	5.0%	10.1%	2.6%	87.4 m	2.8%	6.8%	1.3%
NaVila [45]	69.8 m	7.6%	14.5%	4.1%	80.4 m	4.2%	9.0%	2.0%
OpenFly [28]	61.9 m	10.1%	18.2%	5.8%	73.8 m	5.8%	11.6%	2.9%
Ours	47.4 m	30.1%	31.6%	23.7%	56.4 m	26.4%	28.3%	17.5%

Table 4
Results on *test-long* (seen/unseen).

Method	test-seen				test-unseen			
	NE↓	SR↑	OSR↑	SPL↑	NE↓	SR↑	OSR↑	SPL↑
Random	272.9 m	0.0%	0.0%	0.0%	238.1 m	0.0%	0.0%	0.0%
Seq2Seq	223.7 m	0.8%	2.1%	0.4%	234.6 m	0.4%	1.3%	0.2%
CMA	204.3 m	2.4%	6.6%	1.2%	217.5 m	1.3%	4.3%	0.6%
AerialVLN	189.6 m	3.6%	9.6%	1.8%	204.4 m	2.0%	6.3%	0.8%
Navid	197.2 m	3.0%	8.1%	1.5%	211.6 m	1.7%	5.5%	0.7%
NaVila	184.8 m	4.4%	11.6%	2.2%	199.7 m	2.5%	7.6%	1.0%
OpenFly	169.5 m	6.2%	15.6%	3.2%	187.8 m	3.6%	10.1%	1.4%
Ours	150.7 m	17.9%	22.2%	10.5%	165.4 m	13.8%	17.0%	7.2%

Table 5
Results on *test-reality-unseen*.

Method	NE↓	SR↑	OSR↑	SPL↑
Random	99.6 m	0.0%	0.0%	0.0%
Seq2Seq	86.4 m	1.3%	2.6%	0.7%
CMA	74.8 m	4.1%	8.8%	2.2%
AerialVLN	68.3 m	6.2%	12.9%	3.6%
Navid	72.1 m	5.4%	11.2%	3.0%
NaVila	66.9 m	7.8%	15.6%	4.3%
OpenFly	58.7 m	11.2%	20.4%	6.9%
Ours	37.9 m	25.9%	27.1%	19.4%

trend but with a relatively controlled degradation, indicating certain cross-scene generalization ability. Furthermore, on *test-reality-unseen* (Table 5), where the visual style shift and noise are closer to real deployment, the model requires stronger semantic alignment and robust decision-making; the results show that our method can still achieve good performance under this setting, validating its effectiveness and practical potential under more realistic distributions.

7.3. Effectiveness of landmarks and signmarks

To validate the effectiveness and applicability of signmarks in navigation instructions, we conduct comparative experiments against traditional landmark-based descriptions. Landmark follows OpenFly-style prompt construction and provides more fine-grained natural-language descriptions of visible landmarks along both sides of the road (e.g., appearance, color, and relative position). Signmark instead generates instructions primarily based on signmarks in the scene. Considering that real road environments may have sparse or occluded signmark, we further construct a hybrid prompt (Landmark+Signmark): when signmark cues are insufficient, landmark information is used as a supplement to improve instruction completeness and robustness.

We evaluate on *test-short* and *test-reality-unseen*, covering short-range and more deployment-like settings, respectively, and report results in Table 6.

From Table 6, Signmark performs better than Landmark on both splits, indicating that instructions based on signmarks can provide more stable and identifiable navigation constraints, thus improving success rate and path efficiency. The hybrid strategy Landmark+Signmark brings additional gains in some cases, especially on segments where signmark is sparse, where landmark descriptions can supplement available cues and improve robustness. Meanwhile, in the Signmark-dominant setting, we can significantly reduce lengthy landmark details (e.g., color and material), making instructions closer to everyday human expressions while improving conciseness and usability without sacrificing performance.

7.4. Ablation experiment

We conduct ablation experiments on the signmark-guided agent to quantify the contributions of two key designs: Multi-frames and Signmark subtraj. Unlike OpenFly's setting that saves information at "key nodes" (typically turning points), signmarks in road environments may appear at arbitrary locations and is not limited to intersections. Moreover, only detecting and storing cues at turning points may not match real perception processes, since salient signmarks can often be observed from a distance and provide guidance continuously. Therefore, under a more general continuous-observation setting, we systematically evaluate the gains brought by multi-frame information and historical signmark cues for navigation.

We consider four configurations and report results in Table 7.

As shown in Table 7, multi-frame input brings significant gains over single-frame input on both splits, indicating that temporally continuous visual context strengthens the recognition and disambiguation of signmark and its spatial relations, thereby reducing error and improving success and efficiency. Meanwhile, introducing Signmark subtraj also yields consistent improvements, suggesting that explicitly retaining historical signmark cues as sub-trajectories helps maintain consistent navigation constraints under occlusions, viewpoint changes, or sparse-signmark segments. Overall, Multi frames and Signmark subtraj are complementary, and their combination achieves the best overall performance.

7.5. Qualitative examples

As shown in Fig. 6, we provide several successful examples in both 3D Map and 3DGS environments. The agent can stably capture and recognize sīgnmark during navigation, align it with key trigger conditions in the instruction, and thus make consistent and correct action

Table 6
Effectiveness study of landmark and signmark prompts.

Reference	test-short				test-reality-unseen			
	NE↓	SR↑	OSR↑	SPL↑	NE↓	SR↑	OSR↑	SPL↑
Landmark	55.8 m	26.0%	28.1%	18.9%	41.5 m	23.7%	25.0%	18.2%
Signmark	51.9 m	28.3%	30.0%	20.6%	37.9 m	25.9%	27.1%	19.4%
Land & Signmark	49.6 m	29.4%	31.2%	21.3%	36.8 m	26.6%	28.0%	19.9%

Table 7
Ablation study on *test-short* and *test-reality-unseen*.

Method	test-short				test-reality-unseen			
	NE↓	SR↑	OSR↑	SPL↑	NE↓	SR↑	OSR↑	SPL↑
Single frame & No history	69.2 m	18.6%	20.4%	13.0%	50.1 m	17.0%	18.2%	12.4%
Multi frames & No history	66.8 m	19.4%	21.3%	13.4%	48.7 m	17.6%	19.0%	12.7%
Single frame & Signmark subtraj	54.0 m	26.9%	28.4%	19.1%	40.1 m	24.7%	26.0%	18.5%
Multi frames & Signmark subtraj	51.9 m	28.3%	30.0%	20.6%	37.9 m	25.9%	27.1%	19.4%

Fig. 6. Qualitative navigation examples in 3D Map and 3DGS. We visualize two successful episodes with (left) the trajectory on the top-down map and (right) representative first-person views at key steps (A–D); the agent follows signmark-triggered instructions (e.g., go straight, turn, and stop) to reach the goal in both environments.

decisions (e.g., going straight, turning, and stopping upon arrival), ultimately stopping at the specified goal. These examples indicate that sīgnmark can serve as clear and verifiable navigation anchors, enabling the agent to continuously verify its progress on long road segments and reduce the accumulation of errors caused by misjudgment and drift, thereby improving trajectory execution reliability and termination accuracy. Moreover, in the more deployment-like 3DGS setting, the agent can still complete the task under viewpoint changes and background interference, demonstrating good stability and practical applicability.

8. Concluding remarks and future work

8.1. Conclusion

This paper focuses on the low-altitude urban street scenario, which is closer to practical applications for drone navigation, and proposes the visual-language navigation task SG-VLN centered on street-scene sīgnmarks, along with the construction of a unified platform for high-fidelity simulation and data generation. On one hand, the platform uses UE5 Cesium to realize realistic urban street structures and high-density readable sīgnmark; on the other hand, it introduces 3DGS scenes for cross-domain evaluation that better approximates real imaging distributions, thereby providing a reproducible and scalable experimental platform for sīgnmark-driven navigation under "low-altitude flight with no map/weak localization" conditions. Based on this platform, this paper establishes a SG-VLN dataset covering both short- and long-range navigation, forming a systematic evaluation framework from simple to challenging scenarios. Addressing the key challenges of SG-VLN, this paper proposes the signmark-guided agent model, which uses sīgnmarks as intermediate anchors to drive decision-making, and enhances stability and robustness in complex street scenes through three mechanisms: colloquial instruction rewrite, multi-frame visual fusion, and fine-grained sub-trajectory modeling.

Experimental results show that the proposed method achieves superior navigation performance across different splits (short/long, seen/unseen) and realistic style 3DGS scenarios, particularly in long-range tasks where it can more effectively mitigate cumulative errors. The comparison between landmarks and sīgnmark markers further confirms that sīgnmark markers are denser and easier to verify in urban street scenes, providing navigation constraints in a manner more consistent with human expression habits. Ablation studies also demonstrate that Multi-frames input and Signmark subtraj make a clear contribution to performance improvement.

8.2. Limitations and future work

While our SG-VLN framework demonstrates promising capabilities, several limitations present opportunities for future research. First, scaling up the dataset currently relies on simulation sampling. To enhance environmental diversity without prohibitive manual collection costs, future extensions could incorporate layout-controllable multi-modal diffusion models [46] and multi-view consistent generation [47] to synthesize novel signmarks. Furthermore, generative architectures driven by positive-incentive noise [48] and rectified noise [49] can be applied to prevent mode collapse, ensuring that the synthesized urban scenes maintain physical realism.

At the algorithmic level, complex street backgrounds often induce perceptual noise. Introducing variational positive-incentive noise [50] and estimating beneficial noise through data augmentation [51] or graph augmentation [52] can further explicitly enhance vision-language alignment [53]. Investigating how to optimally inject such beneficial noise into MLLMs [54] will be a crucial step in improving the agent's robustness against low-altitude visual interference. Finally, real-world UAVs must continuously adapt to new cities with unfamiliar signmark styles. To address this, exploring remote-sensing incremental object detection with scale-decoupled topology alignment and pseudo-label refinement [55], together with mixture of noise techniques [56] and multimodal continual learning from multi-scenario perspectives [57], will allow the agent to incrementally acquire signmark recognition and navigation knowledge without catastrophic forgetting of previous environments.

CRediT authorship contribution statement

Yuan Jiang: Writing – review & editing, Writing – original draft, Visualization, Methodology. **Chuang Yang:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Jiaze Li:** Writing – original draft, Methodology. **Cong Zhang:** Methodology, Formal analysis. **Zhigang Yang:** Formal analysis. **Qi Wang:** Visualization, Supervision, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The dataset currently hosted on the Hugging Face repository is the preliminary test dataset adopted for our vision-language navigation experiments. The complete official dataset will be supplemented and updated continuously within the same repository in the subsequent period.

All usable experimental data supporting the findings of this study can be accessed permanently via the following URL:

https://huggingface.co/datasets/sosososogoodman/lovening/tree/mainThe dataset is stored as split TFRecord files, which include scene visual observations, natural language navigation instructions and ground-truth navigation trajectories. No confidential or restricted data was utilized throughout this research work.

References

[1] P. Anderson, Q. Wu, D. Teney, J. Bruce, M. Johnson, N. Sünderhauf, I. Reid, S. Gould, A. Van Den Hengel, Vision-and-language navigation: Interpreting visually-grounded navigation instructions in real environments, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2018, pp. 3674–3683.

[2] A. Ku, P. Anderson, R. Patel, E. Ie, J. Baldridge, Room-across-room: Multilingual vision-and-language navigation with dense spatiotemporal grounding, in: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing, EMNLP, Association for Computational Linguistics, 2020, pp. 4392–4412, http://dx.doi.org/10.18653/v1/2020.emnlp-main.356.

[3] J. Krantz, E. Wijmans, A. Majumdar, D. Batra, S. Lee, Beyond the nav-graph: Vision-and-language navigation in continuous environments, in: European Conference on Computer Vision, Springer, 2020, pp. 104–120.

[4] H. Chen, A. Suhr, D. Misra, N. Snavely, Y. Artzi, Touchdown: Natural language navigation and spatial reasoning in visual street environments, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2019, pp. 12538–12547.

[5] O. Sautenkov, A. Akhmetkazy, Y. Yaqoot, M.A. Mustafa, G. Tadevosyan, A. Lykov, D. Tsetserukou, UAV-VLPA*: A vision-language-path-action system for optimal route generation on a large scales, 2025, arXiv preprint arXiv:2503.02454.

[6] K. Chi, W. Jing, J. Li, Q. Li, Q. Wang, Cross-modal spherical aggregation for weakly supervised remote sensing shadow removal, IEEE Trans. Multimed. (2025).

[7] K. Chi, J. Li, W. Jing, Q. Li, Q. Wang, Neural implicit Fourier transform for remote sensing shadow removal, IEEE Trans. Geosci. Remote Sens. 62 (2024) 1–10.

[8] F. Lian, B. Li, D. Du, H. Zhu, Q. Yang, A reinforcement learning framework for energy-optimal UAV path planning in wind fields, Pattern Recognit. (2025) 112912.

[9] X. Han, J. Gao, C. Yang, Y. Yuan, Q. Wang, Real-time text detection with similar mask in traffic, industrial, and natural scenes, IEEE Trans. Intell. Transp. Syst. 26 (1) (2024) 865–877.

[10] X. Wang, C. Li, L. Yu, L. Han, X. Deng, E. Yang, P. Ren, UAV first view landmark localization with active reinforcement learning, Pattern Recognit. Lett. 125 (2019) 549–555.

[11] J. Ye, S. Papaioannou, P. Kolios, Vlm-rrt: Vision language model guided rrt search for autonomous uav navigation, in: 2025 International Conference on Unmanned Aircraft Systems, ICUAS, IEEE, 2025, pp. 633–640.

[12] T. Wang, Z. Zheng, Y. Sun, C. Yan, Y. Yang, T.-S. Chua, Multiple-environment self-adaptive network for aerial-view geo-localization, Pattern Recognit. 152 (2024) 110363.

[13] S. Liu, H. Zhang, Y. Qi, P. Wang, Y. Zhang, Q. Wu, Aerialvln: Vision-and-language navigation for uavs, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 15384–15394.

[14] Z. Wei, B. Lin, Y. Nie, J. Chen, S. Ma, H. Xu, X. Liang, Unseen from seen: Rewriting observation-instruction using foundation models for augmenting vision-language navigation, IEEE Trans. Neural Netw. Learn. Syst. (2025).

[15] Y. Liu, F. Yao, Y. Yue, G. Xu, X. Sun, K. Fu, Navagent: Multi-scale urban street view fusion for uav embodied vision-and-language navigation, 2024, arXiv preprint arXiv:2411.08579.

[16] J. Chen, B. Lin, X. Liu, L. Ma, X. Liang, K.-Y.K. Wong, Affordances-oriented planning using foundation models for continuous vision-language navigation, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 39, 2025, pp. 23568–23576.

[17] C. Yang, K. Zhuang, M. Chen, H. Ma, X. Han, T. Han, C. Guo, H. Han, B. Zhao, Q. Wang, Traffic sign interpretation via natural language description, IEEE Trans. Intell. Transp. Syst. (2024).

[18] C. Yang, X. Han, T. Han, Y. Su, J. Gao, H. Zhang, Y. Wang, L.-P. Chau, Signeye: Traffic sign interpretation from vehicle first-person view, IEEE Trans. Intell. Transp. Syst. (2025).

[19] J. Lee, T. Miyanishi, S. Kurita, K. Sakamoto, D. Azuma, Y. Matsuo, N. Inoue, CityNav: A large-scale dataset for real-world aerial navigation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, ICCV, IEEE, 2025, pp. 5912–5922, http://dx.doi.org/10.1109/ICCV51701.2025.00559.

[20] Epic Games, City sample project unreal engine demonstration, 2025, Unreal Engine 5.6 Documentation, Epic Developer Community. URL https://dev.epicgames.com/documentation/en-us/unreal-engine/city-sample-project-unreal-engine-demonstration.

[21] Y. Du, T. Fu, Z. Chen, B. Li, S. Su, Z. Zhao, C. Wang, Vl-nav: Real-time vision-language navigation with spatial reasoning, 2025, arXiv preprint arXiv:2502.00931.

[22] D. Nie, X. Guo, Y. Duan, R. Zhang, L. Chen, WMNav: Integrating vision-language models into world models for object goal navigation, in: IEEE/RSJ International Conference on Intelligent Robots and Systems, IROS, IEEE, 2025, pp. 2392–2399, http://dx.doi.org/10.1109/IROS60139.2025.11246684.

[23] Y. Xu, Y. Pan, Z. Liu, H. Wang, Flame: Learning to navigate with multimodal llm in urban environments, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 39, 2025, pp. 9005–9013.
[24] X. Song, W. Chen, Y. Liu, W. Chen, G. Li, L. Lin, Towards long-horizon vision-language navigation: Platform, benchmark and method, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025, pp. 12078–12088.
[25] H. Cai, J. Dong, J. Tan, J. Deng, S. Li, Z. Gao, H. Wang, Z. Su, A. Sumalee, R. Zhong, FlightGPT: Towards generalizable and interpretable uav vision-and-language navigation with vision-language models, in: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, EMNLP, Association for Computational Linguistics, 2025, pp. 6659–6676, http://dx.doi.org/10.18653/v1/2025.emnlp-main.338.
[26] Y. Su, D. An, K. Chen, W. Yu, B. Ning, Y. Ling, Y. Huang, L. Wang, Learning fine-grained alignment for aerial vision-dialog navigation, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 39, 2025, pp. 7060–7068.
[27] E. Michaelsen, J. Meidow, Stochastic reasoning for structural pattern recognition: An example from image-based UAV navigation, Pattern Recognit. 47 (8) (2014) 2732–2744.
[28] Y. Gao, C. Li, Z. You, J. Liu, Z. Li, P. Chen, Q. Chen, Z. Tang, L. Wang, P. Yang, et al., OpenFly: A comprehensive platform for aerial vision-language navigation, 2025, arXiv preprint arXiv:2502.18041.
[29] C. Yang, X. Han, T. Han, H. Han, B. Zhao, Q. Wang, Edge approximation text detector, IEEE Trans. Circuits Syst. Video Technol. (2025).
[30] C. Yang, M. Chen, Y. Yuan, Q. Wang, Reinforcement shrink-mask for text detection, IEEE Trans. Multimed. 25 (2022) 6458–6470.
[31] Z. Yang, W. Zhang, Q. Li, W. Ni, J. Wu, Q. Wang, C^2net: Road extraction via context perception and cross spatial-scale feature interaction, IEEE Trans. Geosci. Remote Sens. 62 (2024) 1–11.
[32] Y. Zhang, J. Zhang, G. Wang, J. Deng, H. Sheng, Y. Muhammad, S. Wei, PolyBuild: An end-to-end method for polygonal building contour extraction from high-resolution remote sensing images, IEEE J. Sel. Top. Appl. Earth Obs. Remote. Sens. (2025).
[33] Cosys-Lab, Cosys-AirSim (GitHub repository). URL https://github.com/Cosys-Lab/Cosys-AirSim.
[34] Z. Lan, Y. Jiang, R. Wang, X. Xie, R. Zhang, Y. Zhu, P. Li, T. Yang, T. Chen, H. Gao, et al., AutoBio: A simulation and benchmark for robotic automation in digital biology laboratory, in: International Conference on Learning Representations, ICLR, 2026.
[35] Z. Yang, D. Zhou, Y. Yang, J. Zhang, Z. Chen, Road extraction from satellite imagery by road context and full-stage feature, IEEE Geosci. Remote. Sens. Lett. 20 (2022) 1–5.
[36] Z. Yang, H. Yao, Q. Li, W. Ni, J. Wu, Q. Wang, Semantic–spatial feature refinement network for road extraction from remote sensing images, IEEE Trans. Geosci. Remote Sens. 64 (2026) 1–10.
[37] Y. Fan, W. Chen, T. Jiang, C. Zhou, Y. Zhang, X. Wang, Aerial vision-and-dialog navigation, in: Findings of the Association for Computational Linguistics: ACL 2023, 2023, pp. 3043–3061.
[38] Y. Qi, Q. Wu, P. Anderson, X. Wang, W.Y. Wang, C. Shen, A.v.d. Hengel, Reverie: Remote embodied visual referring expression in real indoor environments, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 9982–9991.
[39] J. Thomason, M. Murray, M. Cakmak, L. Zettlemoyer, Vision-and-dialog navigation, in: Proceedings of the Conference on Robot Learning (CoRL), in: Proceedings of Machine Learning Research, Vol. 100, PMLR, 2020, pp. 394–406.
[40] D. Misra, A. Bennett, V. Blukis, E. Niklasson, M. Shatkhin, Y. Artzi, Mapping instructions to actions in 3D environments with visual goal prediction, in: Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing, EMNLP, Association for Computational Linguistics, 2018, pp. 2667–2678, http://dx.doi.org/10.18653/v1/D18-1287.
[41] X. Wang, D. Yang, Z. Wang, H. Kwan, J. Chen, W. Wu, H. Li, Y. Liao, S. Liu, Towards realistic UAV vision-language navigation: Platform, benchmark, and methodology, in: International Conference on Learning Representations, ICLR, OpenReview.net, 2025.
[42] C. Yang, H. Ma, X. Han, Y. Yuan, Q. Wang, Text-pass filter: An efficient scene text detector, 2026, arXiv preprint arXiv:2601.18098.
[43] C. Yang, H. Ma, Q. Wang, Instance mask growing on leaf, in: BMVC, 2023, pp. 4–6.
[44] J. Zhang, K. Wang, R. Xu, G. Zhou, Y. Hong, X. Fang, Q. Wu, Z. Zhang, H. Wang, Navid: Video-based VLM plans the next step for vision-and-language navigation, in: Robotics: Science and Systems, RSS, 2024, http://dx.doi.org/10.15607/RSS.2024.XX.079.
[45] A.-C. Cheng, Y. Ji, Z. Yang, X. Zou, J. Kautz, E. Biyik, H. Yin, S. Liu, X. Wang, NaVILA: Legged robot vision-language-action model for navigation, in: RSS, 2025.
[46] S. Huang, S. Huang, P. Luo, H. Zhang, Laytrol: Preserving pretrained knowledge in layout control for multimodal diffusion transformers, 2025, arXiv preprint arXiv:2511.07934.
[47] R. Zhu, Z. Huang, J. Sun, P. Luo, H. Zhang, X. Li, ViewMask-1-to-3: Multi-view consistent image generation via multimodal diffusion models, 2025, arXiv preprint arXiv:2512.14099.
[48] X. Li, Positive-incentive noise, IEEE Trans. Neural Netw. Learn. Syst. 35 (6) (2024) 8708–8714.
[49] Z. Gu, Y. Xu, S. Huang, Y. Guo, H. Zhang, Rectified noise: A generative model using positive-incentive noise, 2025, arXiv preprint arXiv:2511.07911.
[50] H. Zhang, S. Huang, Y. Guo, X. Li, Variational positive-incentive noise: How noise benefits models, IEEE Trans. Pattern Anal. Mach. Intell. (2025).
[51] H. Zhang, Y. Xu, S. Huang, X. Li, Data augmentation of contrastive learning is estimating positive-incentive noise, 2024, arXiv preprint arXiv:2408.09929.
[52] S. Huang, Y. Xu, H. Zhang, X. Li, Learn beneficial noise as graph augmentation, in: Proceedings of the 42nd International Conference on Machine Learning, ICML, 2025.
[53] S. Huang, H. Zhang, X. Li, Enhance vision-language alignment with noise, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 39, 2025, pp. 17449–17457.
[54] R. Zhu, S. Huang, Z. Jiao, H. Zhang, Explore how to inject beneficial noise in MLLMs, 2025, arXiv preprint arXiv:2511.12917.
[55] Y. Zhang, Q. Zhou, J. Gao, Q. Wang, STAR-IOD: Scale-decoupled topology alignment with pseudo-label refinement for remote sensing incremental object detection, ISPRS J. Photogramm. Remote Sens. 238 (2026) 650–663.
[56] K. Jiang, Z. Shi, D. Zhang, H. Zhang, X. Li, Mixture of noise for pre-trained model-based class-incremental learning, 2025, arXiv preprint arXiv:2509.16738.
[57] K. Jiang, S. Huang, X. Chen, J. Shao, H. Zhang, X. Li, Multimodal continual learning with MLLMs from multi-scenario perspectives, 2025, arXiv preprint arXiv:2511.18507.